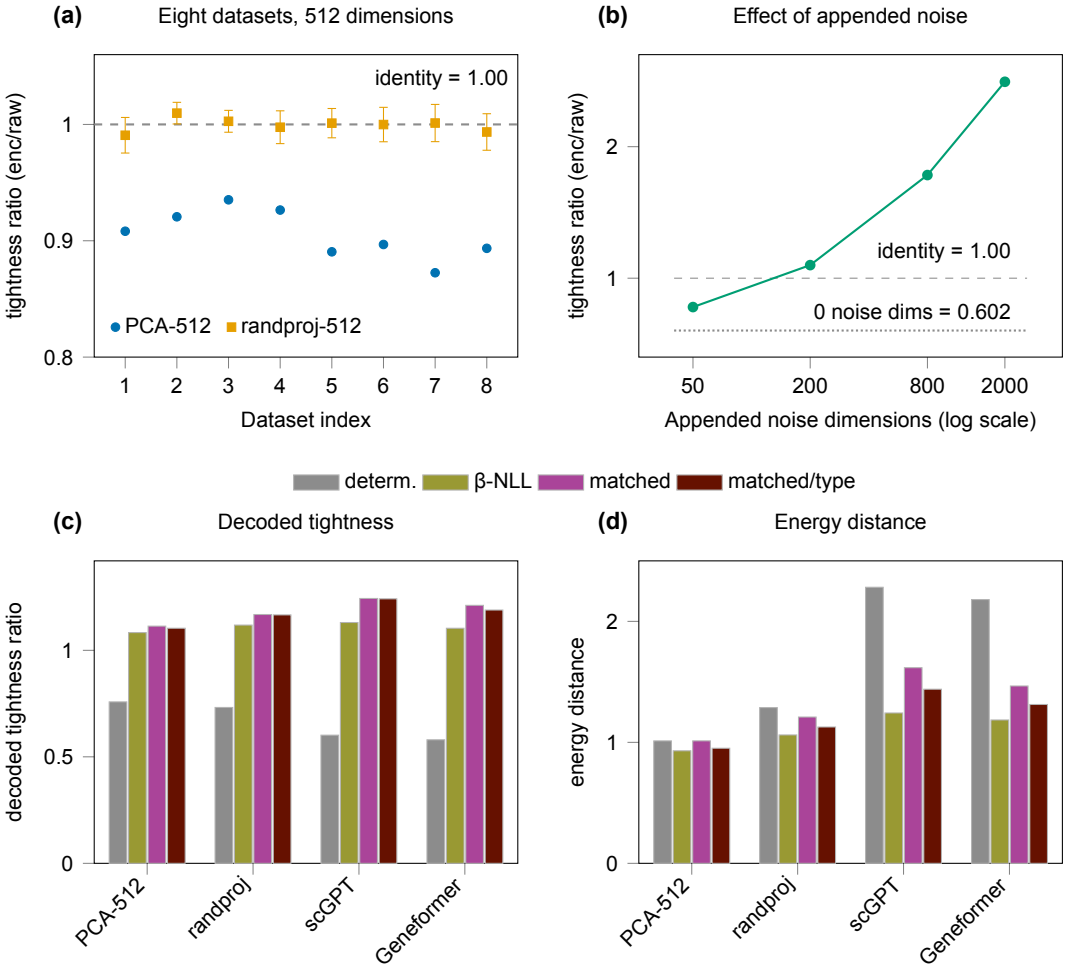
(a)
Eight datasets, 512 dimensions
identity = 1.00
tightness ratio (enc/raw)
1
0.9
0.8
PCA-512
randproj-512
1
2
3
4
5
6
7
8
Dataset index
(b)
Effect of appended noise
tightness ratio (enc/raw)
2
1
identity = 1.00
0 noise dims = 0.602
50
200
800
2000
Appended noise dimensions (log scale)
determ.
β-NLL
matched
matched/type
(c)
Decoded tightness
decoded tightness ratio
1
0.5
0
PCA-512
randproj
scGPT
Geneformer
(d)
Energy distance
energy distance
2
1
0
PCA-512
randproj
scGPT
Geneformer